

Problem set 1

Topic: Linear regression

Due date: 15.06.2026, 23:59

Grading: 30% of your final grade

Notes

- **Report submission:** Submit your answers/reports to Moodle before the deadline. Do not send the report via email. Only the reports submitted to Moodle will be evaluated.
- **Report format:** You need to submit a zip file containing: (1) a report as a PDF document for all problems, and (2) a Jupyter notebook (.ipynb) or similar (e.g. R code or any other program) for Problems 3 & 4. The zip file should be named as “Matriculation number (03xxxxxx), Name”. For example, “03709721, Qinglong Lu”. This information should also be provided at the beginning of the report and the notebook/code. Note that the notebook/code should be submitted with the outputs (i.e., after running each cell). You may write your own code in a programming language of your choice.
- **Report evaluation:** You are evaluated based on your reports and the final answers. So, you have to describe and explain the steps

to solve each problem as clearly as possible. Also, you need to solve these problems individually, and all text needs to be written in your own words. Solutions will be checked for plagiarism and originality. Copying others' solutions will not be tolerated and will result in a fail grade.

1 Problem 1 (10 points)

Suppose you have fitted a multiple linear regression model and the $(X'X)^{-1}$ matrix is:

$$\begin{bmatrix} 0.893758 & -0.0282448 & -0.0175641 \\ -0.0282448 & 0.0013329 & 0.0001547 \\ -0.0175641 & 0.0001547 & 0.009108 \end{bmatrix} \quad (1)$$

1. How many independent variables are in this model? (3 points)
2. If the error sum of squares is 307 and there are 15 observations, what is the estimate of σ^2 ? (3 points)
3. What is the standard error of the regression coefficient $\hat{\beta}_1$? (4 points)

2 Problem 2 (20 points)

Consider the linear regression model

$$y_i = \beta_0 + \beta_1(x_{i1} - \bar{x}_1) + \beta_2(x_{i2} - \bar{x}_2) + \epsilon_i \quad (2)$$

where $\bar{x}_1 = \frac{1}{n} \sum_{i=1}^n x_{i1}$ and $\bar{x}_2 = \frac{1}{n} \sum_{i=1}^n x_{i2}$ and n is the number of observations.

1. Write out the least squares normal equations (i.e. simultaneous equations) for this model. (5 points)

2. Verify that the least squares estimate of the intercept in this model is $\beta_0 = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}$. (5 points)
3. Suppose that we use $y_i - \bar{y}$ as the response variable in this model. What effect will this have on the least squares estimate of the intercept? (5 points)
4. Assuming x_1 and x_2 are positively correlated to each other, please discuss the influence of multi-collinearity on the model. (5 points)

3 Problem 3 (40 points)

The data which can be found in a separate spreadsheet provides the highway gasoline mileage test results for 2005 model year vehicles from DaimlerChrysler. The variable description is written in Table 1.

Table 1: Variable description for Problem 3 data

Variable code	Description
mfr	mfr code
carline	car line name (test vehicle model name)
car/truck	C for passenger vehicle and T for truck
cid	cubic inch displacement of test vehicle
rhp	rated horsepower
trns	transmission code
drv	drive system code
od	overdrive code
etw	equivalent test weight
cmp	compression ratio
axle	axle ratio
n/v	ratio (engine speed versus vehicle speed at 50 mph)
a/c	indicates air conditioning simulation
hc	HC(hydrocarbon emissions) Test level composite results
co	CO(carbon monoxide emissions) Test level composite results
co2	CO2(carbon dioxide emissions) Test level composite results
mpg	fuel economy, miles per gallon

1. Fit a multiple linear regression model to these data to estimate gasoline mileage that uses the following independent variables: cid, rhp, etw, cmp, axle, n/v. (5 points)
2. Estimate σ^2 and the standard errors of the regression coefficients. (5 points)

3. Test for significance of regression using significance level $\alpha = 0.05$. What conclusions can you draw? (5 points)
4. Find the t-test statistic for each independent variable. Using $\alpha = 0.05$, what conclusions can you draw? Does each independent variable contribute to the model? (7 points)
5. Find 99% confidence intervals on the regression coefficients. (6 points)
6. Plot residuals versus $\hat{y}$ and versus each independent variable. Discuss these residual plots. (5 points)
7. Using $\alpha = 0.05$, plot the confidence limits and prediction limits for the mean response. (7 points)

4 Problem 4 (30 points)

The data which can be found in a separate spreadsheet provides the traffic data of a two-lane segment of the SR241-N freeway in California. The data were collected every five minutes. Each observation consists of the flow, average occupancy, and average speed on the segment of interest, the flow ratio between two lanes, and the speed information in the downstream of the segment. The variable description is written in Table 2. This problem asks to predict the average speed of the segment (i.e., variable *speed*) by using the remaining independent variables and their transformations (e.g., x^2).

Table 2: Variable description for Problem 4 data

Variable code	Description
flow	number of vehicles observed within a 5-min interval (veh)
occupancy	the length of time that the detector was occupied by vehicles in this 5-min interval (min)
speed	average speed of all vehicles driving through the detector in this 5-min interval (km/h)
speed_down	average speed of all vehicles driving through the downstream detector in the 5-min interval (km/h)
flow_ratio	the ratio of the flow of target lane over the flow of the adjacent lane

1. Visualise the relationship between every two variables. (5 points)

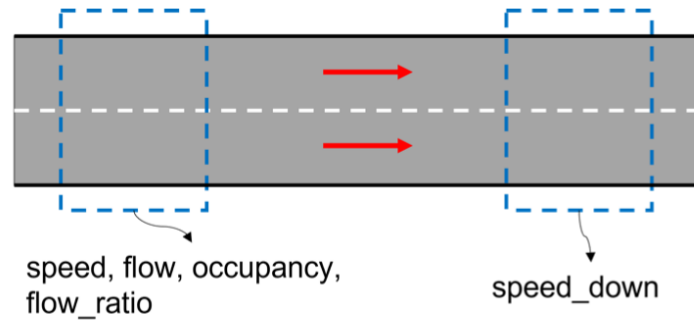


Figure 1: Conceptual illustration of the source of different variables.

2. Fit a multiple linear regression model to the original dataset. Does the model suffer a multicollinearity problem? If yes, which variables cause this issue? (5 points)
3. Select suitable independent variables for the regression model. Note, you may need to do some transformations on the original variables. Explain why you do these transformations, and why you choose these variables? (10 points)
4. Apply linear regression to estimate the relationship between the average speed of the segment and the independent variables you selected in 3. Then apply ridge regression and lasso regression to estimate the relationship between the average speed of the segment and the original set of variables. Compare the results of different models and verify the variables you selected in 3. (10 points)